Spring 2026 Quarterly Report

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Quarterly Report, Spring 2026

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Abstract

The following report recounts the research progress I have made over the course of spring quarter 2026. The report begins with an overview of the research done during winter quarter and discusses the additions and enhancements made. It then discusses the new datasets that were incorporated from the Wind spacecraft and details the integration, masking, and averaging methods used to process the data for analysis. A new technique was also performed: canonical correlation analysis (CCA). The main milestones of this research thus far includes a CCA of IMF B_x, B_y, and B_z with polar rain flux (PRF) from 2010–2015; creating direct scatter plots between Wind measured strahl flux and DMSP measured PRF separated by strahl and B_x orientation from 2003–2014; and conducting a CCA with PRF, strahl flux, B_x, B_y, B_z from 2003–2014. The preliminary conclusions of this analysis are that the most favorable conditions for PRF in the northern hemisphere are parallel strahl, $B_x < 0$, and $B_y > 0$; the most favorable conditions for PRF in the southern hemisphere are antiparallel strahl, $B_x > 0$, and $B_y < 0$; and that strahl orientation has a more significant impact on PRF than IMF orientation.

Introduction

The research this quarter has expanded on the process of analyzing data from the Defense Meteorological Satellite Program (DMSP) by incorporating new evaluation techniques and sources of data. Previously, I have processed differential energy fluxes from DMSP's Special Sensor J5 (SSJ/5) into integrated electron number fluxes and filtered the data strictly for the polar cap. The range of integration was split into three bins: high energy flux (1.39 keV to 30 keV) polar rain flux (PRF) (65 to 949 eV), and total energy flux (30 eV to 30 keV). The filtering process was also done for interplanetary magnetic field (IMF) data from NASA's OMNIWeb database, and each parameter was plotted against each other to search for trends. Trends were modest at best, which incentivised my decision to incorporate more advanced analysis techniques and different datasets.

This quarter, I have conducted a canonical correlation analysis (CCA) to further identify potential trends in the data as well as processing data from the Wind spacecraft's Solar Wind Experiment (SWE) instrument. Differential flux distributions from Wind were integrated over different pitch angles and plotted against fluxes derived from DMSP's SSJ/5. A CCA was also conducted with the two and IMF values from OMNI. From the analysis, I have found that the most favorable conditions for Northern Hemispheric PRF is parallel strahl, $B_x < 0$, and $B_y > 0$ and that the most favorable conditions for Southern Hemispheric PRF is antiparallel strahl, $B_x > 0$, and $B_y < 0$. I have also found that PRF is more dependent on strahl orientation than $B_{x/y}$ orientation.

Methodology

DMSP is a constellation of sun-synchronous polar orbiting satellites at 830 km altitude with a 101 minute orbital period. The primary instrument used for this study is its SSJ/5, a pair of nested triquadrispherical electrostatic analyzers providing electron and ion flux in 19 energy channels from 30 eV to 30 keV (Anderson & Bukowski, 2024). The data is presented in Common Data Format (CDF) available on the NASA Coordinated Data Analysis Web (CDAWeb) database which was used in this study to obtain datasets. It is important to remark that the electron population in that region of the ionosphere is relatively isotropic in pitch angle.

Wind is a spin-stabilized spacecraft with a three second spin period at the Sun-Earth Lagrange position L1 observing solar wind. For this study, Wind's SWE instrument was used. It includes a truncated toroidal electrostatic analyzer called the Electron Strahl Detector and a set of three small electrostatic analyzers called the Vector Electron/Ion Spectrometer designed to determine the ion and electron distribution functions (Wilson III, 2016). The data is also available in CDF format on the NASA CDAWeb database and is sorted into pitch angle bins of 6° width.

An auroral oval boundary detection algorithm by Kilcommons et al. (2017) was used to identify and mask the polar cap. The algorithm identifies ideal polar passes where DMSP intersects the auroral oval twice. Each pass has two equatorial side crossings and two polar side crossings which are marked and timestamped as a result of this algorithm. The design uses integrated electron energy flux from 1.3–30 keV to determine crossings and then applies a figure of merit to determine the most likely candidates. For this study, only the polar-side boundaries were used and a 70 second buffer was added to further restrict the data to the cap. The results of this algorithm are shown in Figure 1.

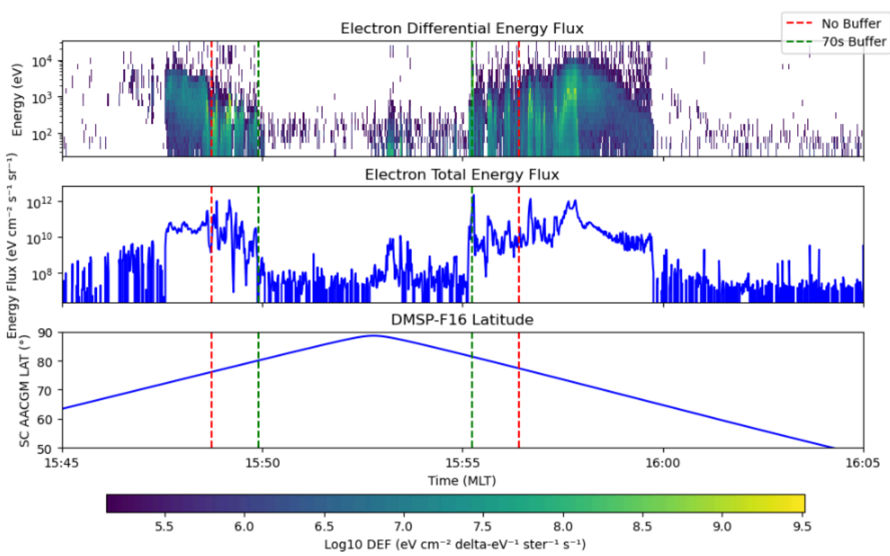

Figure 1. The a polar cap crossing by DMSP F16 showing the original boundaries (red) and 70 second buffer (green)

To conduct the analysis between SSJ/5 and SWE measurements, average differential energy flux was calculated using data from each instrument. For SSJ/5 data, the equation

$$Eq.\ (1)\ \ F\left(E\right) = \frac{\int \Phi\left(E\right) dE}{\int dE}$$

was used over two integration bins, high energy flux (1.39 keV to 30 keV) and PRF (65 to 949 eV). For a non-specific polar pass, $Eq.\ (1)$ yields plots similar to that represented in Figure 2.

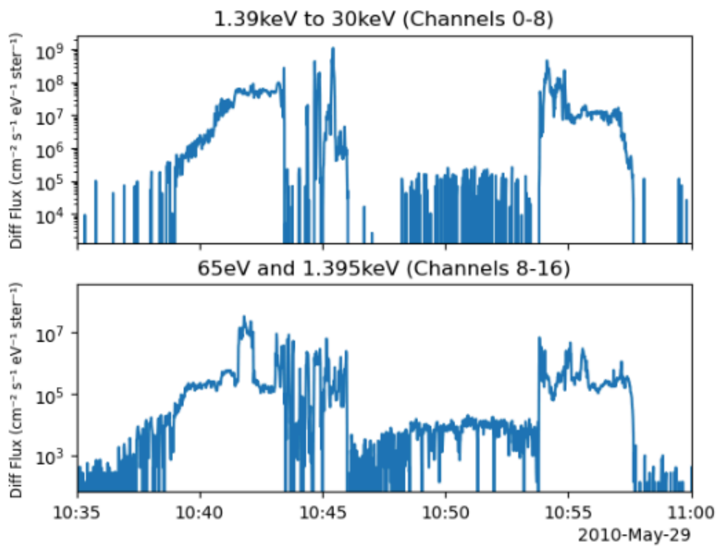

Figure 2. Average differential energy flux calculated using $Eq.\ (1)$ for SWE data over a polar cap pass on May 29th, 2010

What is notable in Figure 2 is the increase in average differential energy flux in the center of the polar cap for the integration zone corresponding to PRF. For SWE, a similar method was used. However, since the provided data preserves pitch angle distribution, the equation

$$Eq.\ (2)\ \ F\left(E\right) = \frac{2\pi \int \Phi\left(\alpha\right) \sin\alpha\, d\alpha}{2\pi \int \sin\alpha\, d\alpha}$$

was used over just one energy channel, 116.1 eV. Three bins of integration were chosen, parallel (0 to 30 degrees), antiparallel (150 to 180 degrees), and intermediate (36 to 144 degrees). The results of this analysis for a non-specific date are shown in Figure 3.

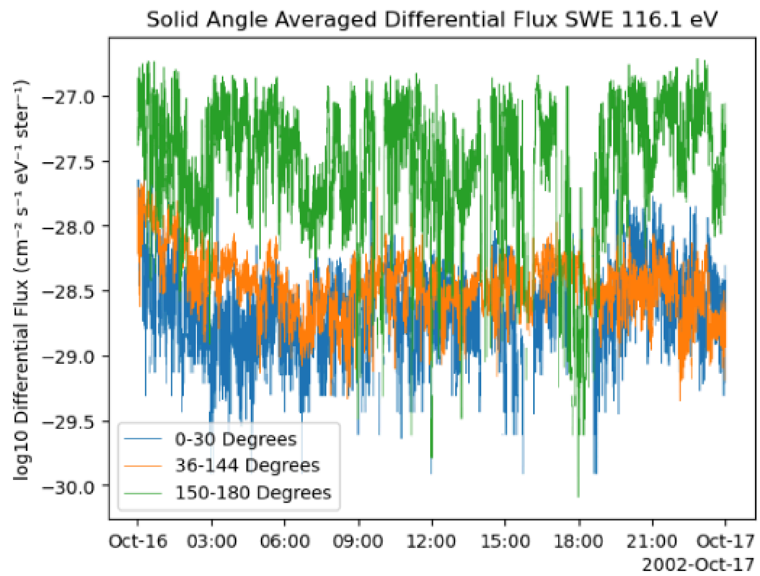

Figure 3. Average differential energy flux calculated using $Eq.\ (2)$ over SWE data on October 17th, 2002

Figure 3 shows that for this range of time at L1, there is a dominance of antiparallel strahl over both parallel strahl and intermediate flux. This is an expected result, as strahl is usually majorly field-aligned in only one orientation. During the course of my analysis, however, I have found regions with a reversal in strahl orientation from parallel to antiparallel dominated and vice versa, corresponding to a satellite current sheet crossing. One of these switch regions is shown in Figure 4.

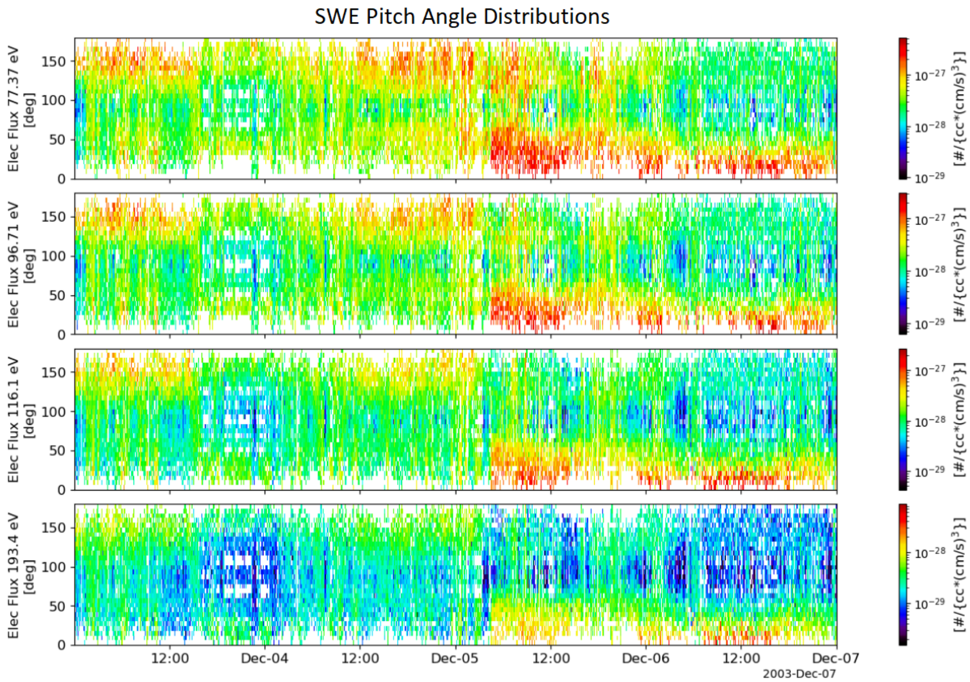

Figure 4. Pitch angle distribution spectrograms for SWE during a current sheet crossing showing a switch from antiparallel to parallel dominated strahl

When conducting the filtering process, the boundaries corresponding to SWE data were shifted to account for the propagation time of electrons from Wind at L1 to DMSP in the ionosphere. The delay was calculated via

$$Eq.\ (3)\ \ 116.1\, eV\ = \frac{1}{2}mv^2$$

for 116.1 eV specifically. $Eq.\ (3)$ was used to determine the electron speed which was then used to determine the time it would take for the measured electrons to travel the minimum 1.5×10^{9} meters to Earth. After integrating and filtering for the polar cap, SSJ/5 flux and SWE flux for each polar cap crossing were collapsed into their respective median value for each crossing. OMNI IMF values were collapsed into the mean value for each crossing period.

To better analyze the processed data for potential correlations, a CCA was applied including SSJ/5 flux, SWE flux, and IMF B_x, B_y, and B_z. This method takes various input variables and forms two composite variables which are linear combinations of those input variables such that the correlation between the two composite variables is maximized. For my analysis, I utilized CCA in a manner more consistent with multiple linear regression rather than a true CCA, since one composite variable contained only SSJ/5 flux while the other contained SWE flux and IMF values; however, the finer details are irrelevant to this study. For more information on using CCA for this type of research, see the study done by Borovsky, (2014).

Results

Before incorporating SWE data into the analysis, a CCA was done on IMF values from OMNI. All input variables span from 2010–2015, and were normalized before the CCA. Multiple sets of input variables were used with sets containing various forms of processed IMF variables. A linear fit was determined for each case and a two-dimensional histogram was also arranged. Set A, containing all IMF values without filtering, "stock" values, is presented in Figure 5.

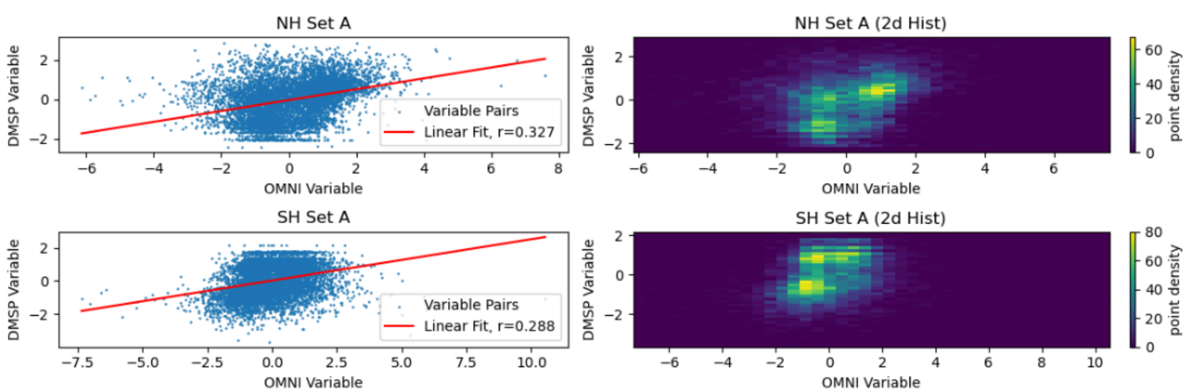

Figure 5. Plots of the composite variables from CCA Set A alongside a two-dimensional histogram.

The results in Figure 5 show a weak correlation between IMF stock values and PRF. The linear fits of r = 0.288 and r = 0.327 are quite low but do exist. What is of further interest is the prominent grouping shown in the two-dimensional histogram, with three distinct clusters of points appearing.

Set C consisted of a set filtered for specific IMF orientation. In the Northern Hemisphere, IMF was filtered for $B_x < 0$, $B_y > 0$, and $B_z < 0$. In the Southern Hemisphere, IMF was filtered for $B_x > 0$, $B_y < 0$, and $B_z < 0$. These were defined as ideal values as opposed to the stock values shown in Figure 5 and were in accordance with the results of Hong et al., (2012). These are shown in Figure 6.

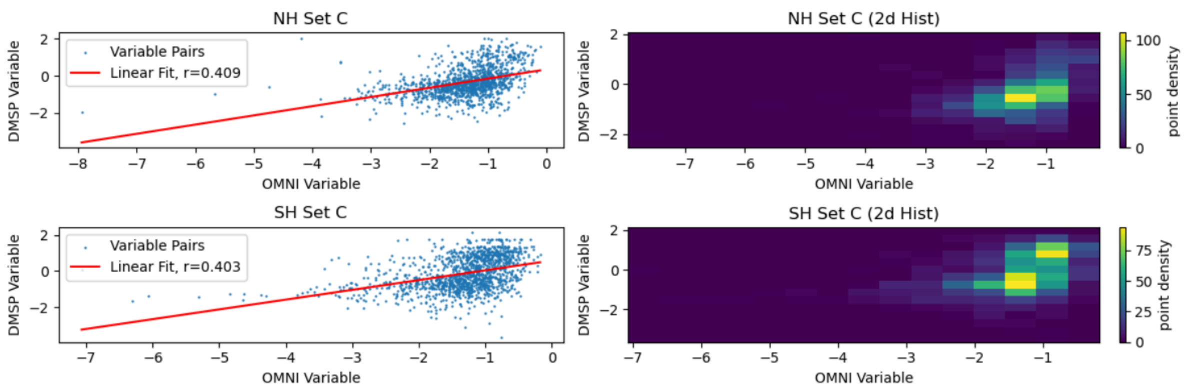

Figure 6. Plots of composite variables of Set C alongside a two-dimentional histogram

Figure 6 shows that there is a sizable increase in correlation coefficients when IMF is filtered by orientation, with linear fits of r = 0.409 and r = 0.403. This indicates that IMF orientation does have some correlation with measured PRF. What is of further note is the loss of clusters in the histogram, which leads to my belief that those clusters correspond to different IMF orientations.

After producing CCA plots with just IMF and PRF, I extended the range of analysis to cover from 2003–2014 and tried a direct comparison between PRF and strahl flux. I created four plots for each hemisphere, with rows corresponding to strahl orientation and columns corresponding to IMF B_x orientation. The results of this analysis for the Northern Hemisphere are shown in Figure 7.

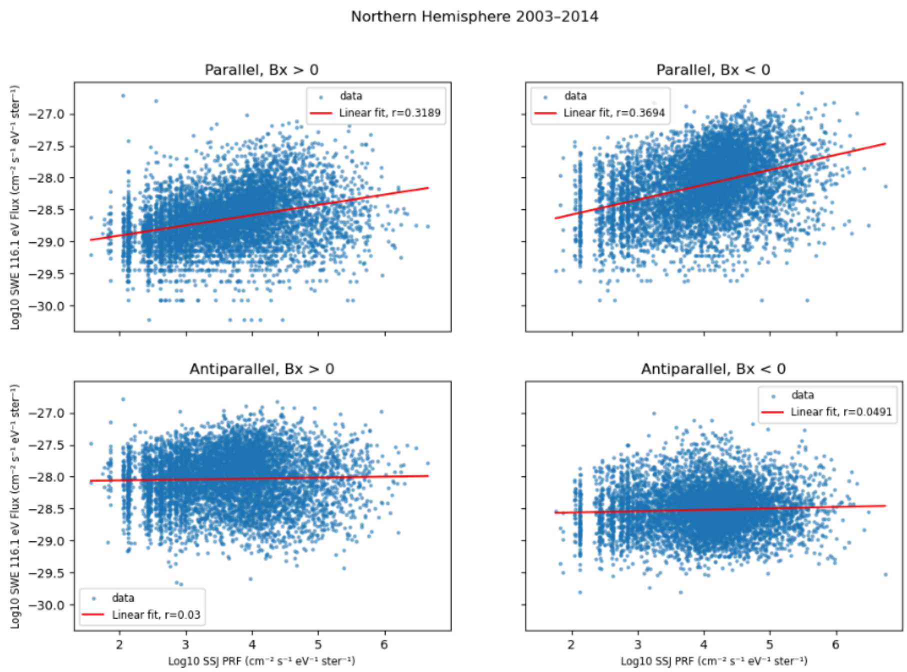

Figure 7. Plots of PRF against strahl flux separated by strahl and IMF orientation in the Northern Hemisphere

Figure 7 shows interesting correlations between strahl orientation, IMF orientation, and PRF. The most prominent aspect of the plots is the relationship of PRF with strahl orientation. Antiparallel strahl appears to have near-zero correlation with Northern Hemisphere PRF with r = 0.0300 and r = 0.0491 whereas the parallel strahl plots have linear fits with r = 0.319 and r = 0.369. IMF B_x orientation also has a factor in PRF correlation with strahl flux, with $B_x < 0$ having a slightly larger correlation coefficient in both instances of strahl orientation as opposed to $B_x > 0$. This indicates that the most favorable conditions for Northern Hemisphere PRF flux is parallel strahl and $B_x < 0$ and that PRF has a higher dependence on strahl orientation than IMF B_x orientation. The results for the Southern Hemisphere are shown in Figure 8.

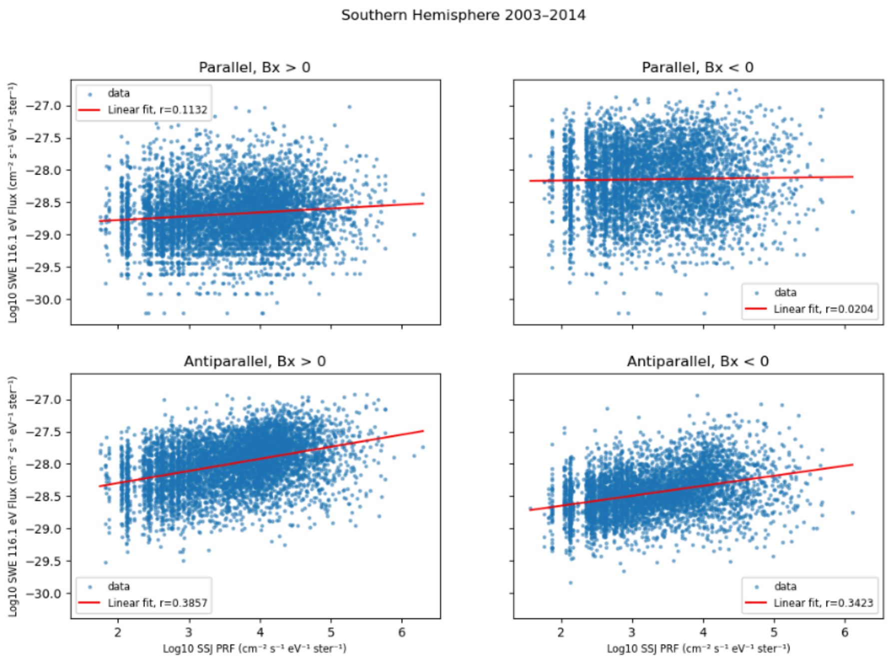

Figure 8. Plots of PRF against strahl flux separated by strahl and IMF orientation in the Southern Hemisphere

The trends in Figure 8 are reversed to what is shown in Figure 7. The large difference in correlation coefficient value between antiparallel and parallel dominated passes and the marginal difference in IMF B_x orientation are present. However, the most favorable conditions for PRF are instead shown to be antiparallel strahl and $B_x > 0$. This indicates that PRF still has a higher dependence on strahl orientation than IMF B_x value in the Southern Hemisphere, but that the direction of that orientation is opposite.

A CCA was also performed between PRF, IMF values, and strahl flux. One composite variable contained only PRF and the other contained B_x, B_y, and strahl flux. Before running this analysis, the latter composite variable was checked for internal correlation via the correlation matrix. The correlation matrix for the Northern Hemisphere is depicted in Table 1 and for the Southern Hemisphere in Table 2.

NH	B_x	B_y	Parallel Strahl	Antiparallel Strahl
B_x	1	-0.423	-0.513	0.561
B_y	-0.423	1	0.384	-0.444
Parallel Strahl	-0.513	0.384	1	-0.210
Antiparallel Strahl	0.561	-0.444	-0.210	1

Table 1. The correlation matrix for the Northern Hemisphere IMF and strahl input variables

SH	B_x	B_y	Parallel Strahl	Antiparallel Strahl
B_x	1	-0.428	-0.500	0.549
B_y	-0.428	1	0.409	-0.422
Parallel Strahl	-0.500	0.409	1	-0.171
Antiparallel Strahl	0.549	-0.422	-0.171	1

Table 2. The correlation matrix for the Southern Hemisphere IMF and strahl input variables

Table 1 and 2 show that there is a moderate correlation between B_x and B_y with parallel and antiparallel strahl. The correlation is not high but is non-negligible, which must be considered when interpreting the results of the CCA. The results for the combined CCA in the Northern Hemisphere are shown in Table 3.

NH Variable	Value
r	0.425
B_x weight	-0.101

B_y weight	0.382
Parallel strahl weight	0.902
Antiparallel strahl weight	0.175

Table 3. Results from the CCA involving PRF, IMF $B_{x/y}$, and parallel/antiparallel strahl flux for the Northern Hemisphere

The correlation coefficient in this instance is a modest r = 0.425 showing a weak to moderate trend. IMF orientation is in accordance with the CCA done in Figure 6 with $B_x < 0$ and $B_y > 0$ being preferred. Parallel strahl is also weighed more than antiparallel strahl. What is most prominent is the degree to which parallel strahl has been weighted by the CCA. A parallel strahl weight of 0.902 far supersedes the other input variables including B_x which supports the trends shown in Figure 7. However, due to the correlation between B_x and strahl of both orientations as shown in Table 1, this large difference in correlation coefficients could be influenced by the incorporation of redundant data. The results of the CCA for the Southern Hemisphere are shown in Table 4.

SH Variable	**Value**
r	0.427
B_x weight	0.044
B_y weight	-0.341
Parallel strahl weight	0.213
Antiparallel strahl weight	0.915

Table 4. Results from the CCA involving PRF, IMF $B_{x/y}$, and parallel/antiparallel strahl flux for the Southern Hemisphere

The correlation coefficient in Table 4 is similar to that in Table 3 with r = 0.427 also indicating a weak to moderate trend. The IMF weights are also in accordance with the CCA done in Figure 6 with $B_x > 0$ and $B_y < 0$ favored. Further, there is a dominance of the antiparallel strahl weight over all other variables (r = 0.915) which is similar to Table 3 having a dominance in parallel strahl weight. Like with the CCA for Table 3, this CCA's corresponding correlation matrix (Table 2) must be considered when interpreting the results, as redundant data may contribute to the higher strahl weight.

Conclusion

To conclude, this quarter has facilitated a large enhancement in what was previously known about the relationship between solar wind and polar rain. The first set of CCAs supported IMF orientation correlations that were indicated by Hong et al. (2012). There exists a noticeable increase in correlation between IMF ideal values and PRF as opposed to stock values, where ideal values include times with $B_x < 0$, $B_y > 0$, and $B_z < 0$ the Northern Hemisphere and $B_x > 0$, $B_y < 0$, and $B_z < 0$ in the Southern Hemisphere. Though not a perfect correlation, these findings support the notion of a Parker Spiral field geometry interacting with the magnetosphere leading to direct field line access during ideal configurations in their respective hemispheres.

A further analysis of strahl flux measured by Wind's SWE instrument uncovered further trends directly between measured strahl flux and PRF. The results shown in Figure 7 and 8 demonstrate a large correlation increase depending on the strahl orientation with parallel strahl having a higher correlation with Northern Hemisphere PRF and antiparallel strahl having a higher correlation with Southern Hemisphere PRF. These results also point to a larger dependence of PRF on strahl orientation rather than B_x orientation. The results are mirrored between the northern and Southern Hemispheres which support Earth's magnetic field geometry. Earth's current magnetic field is oriented such that the field lines point into the Northern Hemisphere and out of the Southern Hemisphere which theoretically imply parallel strahl precipitating in the Northern Hemisphere and antiparallel strahl precipitating in the Southern Hemisphere.

The findings were further supported by a combined IMF and strahl CCA. The CCA produced B_x and B_y weights consistent with their respective orientations produced by the previous CCAs. It also gave weights to strahl flux orientation consistent with what was discovered in Figure 7 and 8. Not only did it extend the trends over a larger time range, it further indicated that strahl orientation was more significant of a PRF indicator as opposed to B_x and B_y orientation. Taking consideration of the inter-correlation between $B_{x/y}$ and strahl, the magnitudes of the weights of ideal parallel and antiparallel strahl fluxes were far greater than that of B_x and B_y in their respective hemispheres.

Though a significant finding, each analysis still showed relatively weak to moderate correlations in every case. It is suspected that by conducting a similar analysis with a closer satellite such as THEMIS-ARTEMIS P1 and P2, which are in lunar orbit as opposed to at L1, these correlations will increase as the local solar wind measured at those points are less likely to miss the Earth entirely on their voyage to the magnetosphere. Though still in its elementary state, this research shows promise in identifying predictors for increased PRF parameters in the polar caps, and it is hoped that further analysis will greatly benefit future ionospheric models.

References

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